Darshan UNIVERSITY
योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 9

Roll No : 514

Name : Meet Indoriya

01) Write a function to calculate BMI given mass and height. (BMI = mass/h**2)

In [7]:
```
def fun_bmi(h, m):
    height_m = h / 100
    bmi = m / (height_m ** 2)
    print("BMI:", round(bmi, 2))

fun_bmi(160, 65)
```

```
BMI: 25.39
```

02) Write a function that add first n numbers.

In [11]:
```
def fun_add(n):
    numberSum = 0
    for i in range(1,n+1,1):
        numberSum += i
    print("Sum: ",numberSum)
fun_add(10)
```

```
Sum:  55
```

03) Write a function that returns 1 if the given number is Prime or 0 otherwise.

In [11]:
```
def isPrime(n):
    primeCount = 1
    for i in range(2,n+1,1):
        if (n % i == 0):
            primeCount += 1
```

```
    if primeCount == 2:
        return 1
    else:
        return 0

isPrime(7)
```

Out[11]: 1

04) Write a function that returns the list of Prime numbers between given two numbers.

In [12]:
```
def PrimeRange(start, end):
    for num in range(start, end + 1):
        if num > 1:
            is_prime = True

            for i in range(2, int(num**0.5) + 1):
                if num % i == 0:
                    is_prime = False
                    break

            if is_prime:
                print(num)

PrimeRange(1, 10)
```

```
2
3
5
7
```

05) Write a function that returns True if the given string is Palindrome or False otherwise.

In [17]:
```
def palindrome(String):
    if String.lower() == String[::-1].lower():
        return True
    else:
        return False
palindrome("Madam")
```

Out[17]: True

06) Write a function that returns the sum of all the elements of the list.

In [23]:
```
def fun_sumElement(*args):
    count2 = 0
    count2 = sum(args)
    print("Sum:",count2)
fun_sumElement(1,2,3,4,5,6,7,8,9,10)
```

```
Sum: 55
```

07) Write a function to calculate the sum of the first element of each tuples inside the list.

In [31]:
```
def listOfTupleSum(tup):

    tuples = list(tup)

    countElement = 0
    for i in tuples:
        countElement += i
    return countElement

tup = (5, 20, 3, 7, 6, 8)
print("Sum of first Elements of Tuple:",listOfTupleSum(tup))
```

```
Sum of first Elements of Tuple: 49
```

08) Write a recursive function to find nth term of Fibonacci Series.

In [5]:
```
def fibonacci(n):
    if n == 0:
        return 0
    elif n == 1:
        return 1
    else:
        return fibonacci(n-1) + fibonacci(n-2)

n = int(input("Enter n: "))
print("Fibonacci term:", fibonacci(n))
```

```
Fibonacci term: 55
```

09) Write a function to get the name of the student based on the given rollno.

Example: Given dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'} find name of student whose rollno = 103

In [57]:
```
dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'}
def elementFinder(ele):
    if ele in dict1:
        return "Found"+str(ele)+", value:"+str(dict1[ele])

    else:
        return "Not Found"
elementFinder(103)
```

Out[57]:
```
'Found103, value:Jay'
```

10) Write a function to get the sum of the scores ending with zero.

Example : scores = [200, 456, 300, 100, 234, 678]

Ans = 200 + 300 + 100 = 600

In [63]:
```
def zero_sumCalculator(l):
    countSum1 = 0
    for i in range(0,len(l),1):
        if l[i] % 10 == 0:
            countSum1 += l[i]
    print(countSum1)

scores = [200, 456, 300, 100, 234, 678]
zero_sumCalculator(scores)
```

```
600
```

11) Write a function to invert a given Dictionary.

hint: keys to values & values to keys

Before : {'a': 10, 'b':20, 'c':30, 'd':40}

After : {10:'a', 20:'b', 30:'c', 40:'d'}

In [68]:
```
def revertPair(di):
    di = { i : j  for j,i in di.items()}
    return di

dict2 =  {'a': 10, 'b':20, 'c':30, 'd':40}
revertPair(dict2)
```

Out[68]:
```
{10: 'a', 20: 'b', 30: 'c', 40: 'd'}
```

12) Write a function that returns the number of uppercase and lowercase letters in the given string.

example : Input : s1 = AbcDEfgh ,Ouptput : no_upper = 3, no_lower = 5

In [76]:
```
def StringCounter(String):
    lowerCounter= 0
    UpperCounter= 0
    String = String.swapcase()
    print(String)
    for i in range(0,len(String),1):
        if String[i].lower() in String:
            lowerCounter+=1
        else:
            UpperCounter+=1
    return lowerCounter, UpperCounter

lr, ur =  StringCounter("AbcDEfgh")

print(f"Lower Count: %d Upper Count:%d"%(lr,ur))
```

```
aBCdeFGH
Lower Count: 3 Upper Count:5
```

13) Write a lambda function to get smallest number from the given two numbers.

In [86]:
```
x = lambda a,b : a if a < b else b
print(x(5,6))
```

```
5
```

14) For the given list of names of students, extract the names having more that 7 characters. Use filter().

In [89]:
```
students = ["meet Indoriya","keval","Elizabeth Olsen","Black Panter","No Name"]
li1 = list(filter(lambda name: len(name) > 7, students))

print(li1)
```

```
['Meet Indoriya', 'Elizabeth Olsen', 'Black Panter']
```

15) For the given list of names of students, convert the first letter of all the names into uppercase. use map().

In [92]:
```
new_li = list(map( lambda x: x.capitalize() ,students))
print(new_li)
```

```
['Meet indoriya', 'Keval', 'Elizabeth olsen', 'Black panter', 'No name']
```

16) Write udfs to call the functions with following types of arguments:

1. Positional Arguments
2. Keyword Arguments
3. Default Arguments
4. Variable Legngth Positional(*args) & variable length Keyword Arguments (**kwargs)
5. Keyword-Only & Positional Only Arguments

In [94]:
```
def pos_argument(a,b):
    if a >b :
        return a
    else:
        return b
pos_argument(5,10)
```

Out[94]:
```
10
```

In [13]:
```
def key_argument(name , surname):
    return name , surname

name , surname =  key_argument("Meet","Indoriya")
print(f"Name: {0} \nSurname: {1}",name ,surname)


def greet(name, *, message):
    print(name, message)

greet("Meet", message="Good Morning")


def divide(a, b, /):
```

```
    return a / b

print(divide(10, 2))
```

```
Name: 0
Surname: 1 Meet Indoriya
Meet Good Morning
5.0
```

In [ ]:

```

```